

Stats 596 – Homework 1

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1. **Exercise 1.3.** Let A, B, C , and D be matrices of appropriate dimensions and write

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad \tilde{A} = A - BD^{-1}C.$$

- (a) Show that if D and $\tilde{A}$ are nonsingular, then M is nonsingular and

$$M^{-1} = \begin{pmatrix} \tilde{A}^{-1} & -\tilde{A}^{-1}BD^{-1} \\ -D^{-1}C\tilde{A}^{-1} & D^{-1} + D^{-1}C\tilde{A}^{-1}BD^{-1} \end{pmatrix}.$$

Give a direct proof using the orthogonalization idea or a variation.

- (b) Show that if D and M are nonsingular, then $\tilde{A}$ is nonsingular and the preceding inverse formula holds.
(c) Suppose A, D , and $\tilde{A}$ are nonsingular. Show that

$$\tilde{D} = D - CA^{-1}B$$

is nonsingular and that

$$\tilde{D}^{-1} = D^{-1} + D^{-1}C\tilde{A}^{-1}BD^{-1}.$$

- (d) As an application, show that if $u^T u \neq 1$, then $I - uu^T$ is nonsingular and

$$(I - uu^T)^{-1} = I + (1 - u^T u)^{-1}uu^T.$$

Solution: Take A to be $m \times m$ and D to be $q \times q$. The matrix $\tilde{A} = A - BD^{-1}C$ is the Schur complement of D in M .

(a) Subtracting BD^{-1} times the second block row from the first removes B ; this residualizes A on the D -block, which is the orthogonalization idea. In matrix form,

$$M = LT, \quad L = \begin{pmatrix} I & BD^{-1} \\ 0 & I \end{pmatrix}, \quad T = \begin{pmatrix} \tilde{A} & 0 \\ C & D \end{pmatrix},$$

as can be checked by multiplying out. Both factors are invertible, with

$$L^{-1} = \begin{pmatrix} I & -BD^{-1} \\ 0 & I \end{pmatrix}, \quad T^{-1} = \begin{pmatrix} \tilde{A}^{-1} & 0 \\ -D^{-1}C\tilde{A}^{-1} & D^{-1} \end{pmatrix},$$

again by direct multiplication. So M is invertible, and multiplying out $M^{-1} = T^{-1}L^{-1}$ gives the stated formula.

(b) L is invertible whenever D is, so $T = L^{-1}M$ is invertible. If $\tilde{A}v = 0$ for some $v \neq 0$, put $w = -D^{-1}Cv$; then

$$T \begin{pmatrix} v \\ w \end{pmatrix} = \begin{pmatrix} \tilde{A}v \\ Cv + Dw \end{pmatrix} = 0,$$

contradicting invertibility of T . So $\tilde{A}$ is invertible and part (a) applies.

(c) Let $\tilde{D} = D - CA^{-1}B$. Eliminating with A as the pivot instead gives

$$M = L_A U_A, \quad L_A = \begin{pmatrix} I & 0 \\ CA^{-1} & I \end{pmatrix}, \quad U_A = \begin{pmatrix} A & B \\ 0 & \tilde{D} \end{pmatrix}.$$

Since D and $\tilde{A}$ are nonsingular, which are exactly the hypotheses of (a), M is nonsingular. Therefore $U_A = L_A^{-1}M$ is nonsingular too. A block upper-triangular matrix whose $(1, 1)$ block A is invertible can only be invertible if its $(2, 2)$ block is: if $\tilde{D}v = 0$ with $v \neq 0$, then $w = -A^{-1}Bv$ gives $U_A \begin{pmatrix} w \\ v \end{pmatrix} = 0$. Hence $\tilde{D}$ is invertible, and

$$M^{-1} = U_A^{-1}L_A^{-1} = \begin{pmatrix} A^{-1} + A^{-1}B\tilde{D}^{-1}CA^{-1} & -A^{-1}B\tilde{D}^{-1} \\ -\tilde{D}^{-1}CA^{-1} & \tilde{D}^{-1} \end{pmatrix}.$$

Since M has only one inverse, this $(2, 2)$ block equals the one from (a): $\tilde{D}^{-1} = D^{-1} + D^{-1}C\tilde{A}^{-1}BD^{-1}$.

(d) Apply (c) with $A = 1$, $B = u^\top$, $C = u$, and $D = I$. Then $\tilde{A} = 1 - u^\top u \neq 0$ and $\tilde{D} = I - uu^\top$, so $I - uu^\top$ is invertible and

$$(I - uu^\top)^{-1} = I + u(1 - u^\top u)^{-1}u^\top = I + (1 - u^\top u)^{-1}uu^\top.$$

2. **Exercise 1.5.** Let X be an $n \times p$ covariate matrix and let Y be an $n \times 1$ response vector.

(a) Show that if $n \geq p$ and X has rank p , then $X^\top X$ is nonsingular and

$$\hat{\beta} = (X^\top X)^{-1}X^\top Y$$

is the unique solution of $\min_{\beta \in \mathbb{R}^p} \|Y - X\beta\|_2^2$.

(b) Show that if $n \leq p$ and X has rank n , then $XX^\top$ is nonsingular and

$$\tilde{\beta} = X^\top (XX^\top)^{-1} Y$$

is the unique solution of

$$\min_{\beta \in \mathbb{R}^p} \|\beta\|_2 \quad \text{subject to} \quad Y = X\beta.$$

(c) Show that, regardless of the rank of X ,

$$(X^\top X)^+ X^\top Y = X^\top (XX^\top)^+ Y.$$

Solution: (a) Full column rank means $Xb \neq 0$ whenever $b \neq 0$, so $b^\top X^\top X b = \|Xb\|_2^2 > 0$; thus $X^\top X$ is positive definite and invertible. Let $\hat{r} = Y - X\hat{\beta}$. Then $X^\top \hat{r} = X^\top Y - X^\top X(X^\top X)^{-1} X^\top Y = 0$, so for any $\beta = \hat{\beta} + h$,

$$\|Y - X\beta\|_2^2 = \|\hat{r} - Xh\|_2^2 = \|\hat{r}\|_2^2 + \|Xh\|_2^2,$$

because the cross term $2h^\top X^\top \hat{r}$ vanishes. This is larger than $\|\hat{r}\|_2^2$ unless $Xh = 0$, that is, unless $h = 0$.

(b) $\text{rank}(X) = n$ means $\text{rank}(X^\top) = n$, so $\mathcal{N}(X^\top) = \{0\}$ and $X^\top a \neq 0$ for $a \neq 0$. Thus $a^\top X X^\top a = \|X^\top a\|_2^2 > 0$ and $XX^\top$ is invertible. Write $\tilde{\beta} = X^\top \tilde{a}$ with $\tilde{a} = (XX^\top)^{-1} Y$; then $X\tilde{\beta} = Y$. Any other solution of $X\beta = Y$ has the form $\beta = \tilde{\beta} + h$ with $Xh = 0$, and $h^\top \tilde{\beta} = (Xh)^\top \tilde{a} = 0$. Hence $\|\beta\|_2^2 = \|\tilde{\beta}\|_2^2 + \|h\|_2^2$, which is smallest only when $h = 0$.

(c) Take the SVD $X = U\Sigma V^\top$, where U and V are orthogonal and Σ is $n \times p$ and diagonal, with positive singular values $s_1, \dots, s_r$ followed by zeros. Since U and V are orthogonal, $(X^\top X)^+ = V(\Sigma^\top \Sigma)^+ V^\top$ and $(XX^\top)^+ = U(\Sigma \Sigma^\top)^+ U^\top$, so

$$(X^\top X)^+ X^\top = V(\Sigma^\top \Sigma)^+ \Sigma^\top U^\top, \quad X^\top (XX^\top)^+ = V \Sigma^\top (\Sigma \Sigma^\top)^+ U^\top.$$

Every matrix between V and $U^\top$ is diagonal, so it is enough to compare diagonal entries. For $s_j > 0$ they are $s_j^{-2} s_j = s_j^{-1}$ and $s_j s_j^{-2} = s_j^{-1}$; for $s_j = 0$ both are 0. So the two sides are equal (both are X^+). With full column rank this is the estimator from (a), and with full row rank it is the minimum-norm solution from (b).

3. **Exercise 1.6.** Consider the linear regression model

$$Y_i \sim X_i^\top \beta, \quad X_i = (1, X_{i,1:p}^\top)^\top, \quad i = 1, \dots, n.$$

Let X be the $n \times (p+1)$ design matrix, and suppose $\hat{\beta} = (X^\top X)^{-1} X^\top Y$. Show that $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_{1:p}^\top)^\top$, where

$$\hat{\beta}_0 = \bar{Y} - \bar{X}_{1:p}^\top \hat{\beta}_{1:p}$$

and

$$\hat{\beta}_{1:p} = (\tilde{X}_{1:p}^T \tilde{X}_{1:p})^{-1} \tilde{X}_{1:p}^T Y,$$

with $\tilde{X}_{1:p}$ denoting the matrix of centered covariates.

Solution: Write $X = (\mathbf{1}, X_{1:p})$, and assume X has full column rank (needed for $(X^T X)^{-1}$ to exist). Let $C = I - \frac{1}{n} \mathbf{1} \mathbf{1}^T$ be the centering matrix, so $CY = Y - \mathbf{1} \bar{Y}$ and $\tilde{X}_{1:p} = CX_{1:p} = X_{1:p} - \mathbf{1} \bar{X}_{1:p}^T$. Note that C is symmetric and $C^2 = C$.

Intercept. For fixed b , $\|Y - \mathbf{1}\beta_0 - X_{1:p}b\|_2^2$ is a quadratic in β_0 , and setting its derivative $-2\mathbf{1}^T(Y - \mathbf{1}\beta_0 - X_{1:p}b)$ to zero gives

$$\hat{\beta}_0(b) = \bar{Y} - \bar{X}_{1:p}^T b.$$

Since $\hat{\beta}$ minimizes over (β_0, b) jointly, its intercept must be the best intercept for its own slope, so

$$\hat{\beta}_0 = \bar{Y} - \bar{X}_{1:p}^T \hat{\beta}_{1:p}.$$

Slope. Plugging in $\hat{\beta}_0(b)$, the residual becomes

$$Y - \mathbf{1}\hat{\beta}_0(b) - X_{1:p}b = (Y - \mathbf{1}\bar{Y}) - (X_{1:p} - \mathbf{1}\bar{X}_{1:p}^T)b = CY - \tilde{X}_{1:p}b.$$

So $\hat{\beta}_{1:p}$ minimizes $\|CY - \tilde{X}_{1:p}b\|_2^2$, which is a no-intercept regression of CY on $\tilde{X}_{1:p}$.

This regression has a unique solution because $\tilde{X}_{1:p}$ has full column rank: if $\tilde{X}_{1:p}b = 0$, then $X_{1:p}b = \mathbf{1}(\bar{X}_{1:p}^T b)$ is constant, so $(\mathbf{1}, X_{1:p})(-\bar{X}_{1:p}^T b, b^T)^T = 0$, and full rank of X forces $b = 0$. By Exercise 1.5(a),

$$\hat{\beta}_{1:p} = (\tilde{X}_{1:p}^T \tilde{X}_{1:p})^{-1} \tilde{X}_{1:p}^T CY = (\tilde{X}_{1:p}^T \tilde{X}_{1:p})^{-1} \tilde{X}_{1:p}^T Y,$$

where the last step uses $\tilde{X}_{1:p}^T C = X_{1:p}^T C^2 = \tilde{X}_{1:p}^T$.

4. Exercise 1.7. Consider

$$Y_i \sim X_i^T \beta = Z_i \theta + W_i^T \alpha, \quad X = (Z, W), \quad \hat{\beta} = (\hat{\theta}, \hat{\alpha}^T)^T.$$

Let R_Y be the residual vector from regressing Y on W , and let R_Z be the residual vector from regressing Z on W . Define

$$\hat{\sigma}^2 = \frac{\|Y - X\hat{\beta}\|_2^2}{n - p}, \quad \tilde{\sigma}^2 = \frac{\|R_Y - R_Z\hat{\theta}\|_2^2}{n - 1}.$$

(a) Show that $\tilde{\sigma}^2 = \frac{n-p}{n-1} \hat{\sigma}^2$.

(b) Show that $(R_Z^\top R_Z)^{-1}$ equals the $(1, 1)$ element of $(X^\top X)^{-1}$.

(c) Let $D = \text{diag}(\hat{r}_1^2, \dots, \hat{r}_n^2)$, where $\hat{r}_i = Y_i - X_i^\top \hat{\beta}$. Show that

$$(R_Z^\top R_Z)^{-1} (R_Z^\top D R_Z) (R_Z^\top R_Z)^{-1}$$

equals the $(1, 1)$ element of

$$(X^\top X)^{-1} (X^\top D X) (X^\top X)^{-1}.$$

Solution: Here Z is $n \times 1$, W is $n \times (p - 1)$, and $X = (Z, W)$ has full column rank p . Let $M_W = I - W(W^\top W)^{-1}W^\top$, which is symmetric with $M_W^2 = M_W$; then $R_Y = M_W Y$ and $R_Z = M_W Z$. Note $R_Z \neq 0$, since otherwise $Z \in \mathcal{C}(W)$ and X would not have full rank.

(a) For fixed θ , the best α comes from regressing $Y - Z\theta$ on W , namely $\hat{\alpha}(\theta) = (W^\top W)^{-1}W^\top(Y - Z\theta)$, and the residual is

$$Y - Z\theta - W\hat{\alpha}(\theta) = M_W(Y - Z\theta) = R_Y - R_Z\theta.$$

The joint minimizer must use the best α for its own θ (otherwise replacing $\hat{\alpha}$ by $\hat{\alpha}(\hat{\theta})$ would strictly lower the criterion), so $\hat{\alpha} = \hat{\alpha}(\hat{\theta})$, and $\hat{\theta}$ minimizes $\|R_Y - R_Z\theta\|_2^2$. Hence the two regressions have identical residuals,

$$Y - X\hat{\beta} = R_Y - R_Z\hat{\theta},$$

which is the Frisch–Waugh–Lovell theorem. So both variance estimates have the same numerator $(n - p)\hat{\sigma}^2$, and $\tilde{\sigma}^2 = \frac{n-p}{n-1}\hat{\sigma}^2$.

(b) Partition

$$X^\top X = \begin{pmatrix} Z^\top Z & Z^\top W \\ W^\top Z & W^\top W \end{pmatrix}.$$

Both $X^\top X$ and $W^\top W$ are nonsingular, so by Exercise 1.3 the $(1, 1)$ entry of $(X^\top X)^{-1}$ is the inverse of the Schur complement

$$Z^\top Z - Z^\top W(W^\top W)^{-1}W^\top Z = Z^\top M_W Z = R_Z^\top R_Z.$$

(c) Let $s = R_Z^\top R_Z$. By the block inverse formula in Exercise 1.3, the first column of $(X^\top X)^{-1}$ is

$$q = \begin{pmatrix} s^{-1} \\ -(W^\top W)^{-1}W^\top Z s^{-1} \end{pmatrix},$$

so

$$Xq = s^{-1}(Z - W(W^\top W)^{-1}W^\top Z) = s^{-1}R_Z.$$

Since $(X^\top X)^{-1}$ is symmetric, the $(1, 1)$ entry of the sandwich is

$$q^\top X^\top D X q = (Xq)^\top D (Xq) = (R_Z^\top R_Z)^{-1} (R_Z^\top D R_Z) (R_Z^\top R_Z)^{-1}.$$

By (a) the two regressions have the same residuals, so D is the same matrix in both.

5. **Exercise 1.8.** Generate the data using the following R code:

```

1 library(MASS)
2 set.seed(0)
3
4 n <- 200
5 p <- 4
6 mu <- rep(0,p)
7 msig <- toeplitz(.8^(1:p-1))
8 sig <- .5
9
10 x <- mvrnorm(n, mu, msig)
11 y <- x[,1] + pmax(0, apply(x[,2:p],1,sum))^2 / 3 +
12     rnorm(n, 0, sig)
13 cbind(y,x)[1:3,]

```

- Report the first three rows. Fit a linear regression of y on x with an intercept, and report all coefficients, degrees-of-freedom-adjusted model-based standard errors, and model-robust standard errors.
- Compute the same quantities using residual-on-residual regressions.
- Find the population target values of all coefficients analytically and numerically.

Solution: The covariance matrix is $\Sigma_{jk} = 0.8^{|j-k|}$: in R, $1:p-1$ means $(1:p)-1$, which is $0:3$.

(a) The first three generated rows are

Row	y	x_1	x_2	x_3	x_4
1	-2.191988	-1.479972	-1.698022	-0.465721	-0.818119
2	-0.502206	-0.662950	0.653037	0.738444	0.336596
3	0.302931	-0.386687	-0.993984	-1.748198	-1.520414

Let $X = (\mathbf{1}, x)$, with $n = 200$ rows and $k = 5$ columns, and let $\hat{r} = y - X\hat{\beta}$. The model-based variance is $\hat{\sigma}^2(X^\top X)^{-1}$ with $\hat{\sigma}^2 = \hat{r}^\top \hat{r} / (n - k)$, and the robust (HC0) variance is $(X^\top X)^{-1} X^\top \hat{D} X (X^\top X)^{-1}$ with $\hat{D} = \text{diag}(\hat{r}_1^2, \dots, \hat{r}_n^2)$.

Coefficient	Estimate	Model SE	HC0 SE
Intercept	0.954681	0.098797	0.097826
x_1	0.683716	0.163978	0.142299
x_2	0.739820	0.209943	0.225490
x_3	0.256060	0.220531	0.217935
x_4	0.868421	0.183722	0.235053

(b) For each coefficient j , residualize the column X_j on the other four columns to get R_j . By Exercise 1.7 (which applies to any column after reordering),

$$\hat{\beta}_j = \frac{R_j^\top y}{R_j^\top R_j}, \quad \text{SE}_{\text{mod},j} = \sqrt{\frac{\hat{\sigma}^2}{R_j^\top R_j}}, \quad \text{SE}_{\text{HC0},j} = \sqrt{\frac{R_j^\top \hat{D} R_j}{(R_j^\top R_j)^2}}.$$

Here $\hat{\sigma}^2$ uses the full-model degrees of freedom $n - k = 195$; by Exercise 1.7(a), using $n - 1$ from the residual-on-residual fit would instead shrink each model SE by a factor of $\sqrt{195/199}$. These formulas reproduce the table above, with a maximum absolute difference of 6.44×10^{-15} .

(c) Let $a = (0, 1, 1, 1)^\top$ and $S = a^\top X = X_2 + X_3 + X_4$, so $Y = X_1 + S^2/3 + \varepsilon$ with ε independent of X . The target minimizes $\mathbb{E}(Y - \beta_0 - X^\top b)^2$. Since $\mathbb{E}X = 0$, its normal equations give $\beta_0^* = \mathbb{E}Y$ and $b^* = \Sigma^{-1} \text{Cov}(X, Y)$.

Intercept. $S \sim N(0, v_S)$ with

$$v_S = 3 + 2\{\text{Cov}(X_2, X_3) + \text{Cov}(X_2, X_4) + \text{Cov}(X_3, X_4)\} = 3 + 2(0.8 + 0.64 + 0.8) = 7.48.$$

By symmetry of S , $\mathbb{E}S_+^2 = \frac{1}{2}\mathbb{E}S^2 = v_S/2$, so $\beta_0^* = \mathbb{E}(S_+^2)/3 = v_S/6 = 1.2466667$.

Slopes. $\text{Cov}(X, Y) = \Sigma e_1 + \frac{1}{3}\mathbb{E}(XS_+^2)$. Because (X, S) is jointly normal, $\mathbb{E}(X | S) = (\Sigma a / v_S) S$, so

$$\mathbb{E}(XS_+^2) = \frac{\Sigma a}{v_S} \mathbb{E}(S^3 \mathbf{1}\{S > 0\}) = \frac{\Sigma a}{v_S} v_S^{3/2} \sqrt{2/\pi},$$

using $\mathbb{E}(Z^3 \mathbf{1}\{Z > 0\}) = \sqrt{2/\pi}$ for $Z \sim N(0, 1)$. Therefore

$$b^* = e_1 + \lambda a, \quad \lambda = \frac{\sqrt{v_S}}{3} \sqrt{\frac{2}{\pi}} = 0.7273938,$$

and $\beta^* = (1.2466667, 1, 0.7273938, 0.7273938, 0.7273938)^\top$.

The following R functions implement the direct and residual-on-residual calculations used above.

```

1 ols_stats <- function(x, y) {
2   X <- cbind(1, x)
3   n <- nrow(X)
4   k <- ncol(X)
5   XtX_inv <- solve(crossprod(X))
6   beta_hat <- drop(XtX_inv %*% crossprod(X, y))
7   residual <- drop(y - X %*% beta_hat)
8
9   V_model <- sum(residual^2) / (n - k) * XtX_inv
10  V_robust <- XtX_inv %*%

```

```

11     crossprod(X, X * residual^2) %*% XtX_inv
12
13     list(beta = beta_hat, residual = residual,
14          V_model = V_model, V_robust = V_robust, X = X)
15 }
16
17 ror_stats <- function(X, y) {
18     n <- nrow(X)
19     k <- ncol(X)
20     full <- ols_stats(X[, -1, drop = FALSE], y)
21     out <- matrix(NA_real_, k, 3)
22
23     for (j in seq_len(k)) {
24         Z <- X[, j]
25         W <- X[, -j, drop = FALSE]
26         RZ <- drop(Z - W %*%
27                    solve(crossprod(W), crossprod(W, Z)))
28
29         out[j, 1] <- sum(RZ * y) / sum(RZ^2)
30         out[j, 2] <- sqrt(
31             sum(full$residual^2) / (n - k) / sum(RZ^2))
32         out[j, 3] <- sqrt(
33             sum(RZ^2 * full$residual^2) / sum(RZ^2)^2)
34     }
35     colnames(out) <- c("estimate", "model_se", "robust_se")
36     out
37 }
38
39 fit <- ols_stats(x, y)
40 standard <- cbind(
41     estimate = fit$beta,
42     model_se = sqrt(diag(fit$V_model)),
43     robust_se = sqrt(diag(fit$V_robust)))
44 ror <- ror_stats(fit$X, y)
45 standard
46 ror
47 max(abs(standard - ror))

```

6. **Exercise 1.9.** Using the random-design setting of Exercise 1.8, run 2000 repeated simulations and report:
- (a) Monte Carlo biases and standard deviations of the five estimated coefficients.
 - (b) Square roots of the Monte Carlo means of model-based variances and the 90% and 95% Wald coverage proportions.

- (c) Square roots of the Monte Carlo means of model-robust variances and the 90% and 95% Wald coverage proportions.
- (d) Summarize the conclusions from the simulation.

Solution: The target is $\beta^* = (1.2466667, 1, 0.7273938, 0.7273938, 0.7273938)^\top$ from Exercise 1.8. There are 2000 replications after `set.seed(0)`; each draws new X and Y , and the 90% and 95% Wald intervals use the normal critical values 1.645 and 1.960. “RMV” is the square root of the Monte Carlo mean of the estimated variances.

(a)–(b) Model-based results (2000 replications):

Coefficient	Target	Bias	MC SD	Model RMV	Cov. 90%	Cov. 95%
Intercept	1.246667	-0.016471	0.144232	0.141715	0.8655	0.9310
x_1	1.000000	-0.001998	0.229680	0.237606	0.9130	0.9585
x_2	0.727394	-0.014301	0.315431	0.304206	0.8755	0.9345
x_3	0.727394	0.002095	0.311521	0.303437	0.8885	0.9425
x_4	0.727394	-0.003744	0.253048	0.236669	0.8750	0.9280

(c) HC0 results (same 2000 replications):

Coefficient	HC0 RMV	Cov. 90%	Cov. 95%
Intercept	0.140476	0.8610	0.9280
x_1	0.229364	0.9025	0.9545
x_2	0.307878	0.8795	0.9315
x_3	0.309789	0.8830	0.9385
x_4	0.250629	0.8805	0.9295

(d) The linear model is misspecified, since the true mean contains $S_+^2/3$, but OLS still estimates the projection target β^* .

Bias. The biases are small compared with the MC SDs. The intercept bias (-0.0165) is about five Monte Carlo standard errors ($0.144/\sqrt{2000} \approx 0.0032$) from zero, which is strong Monte Carlo evidence of finite-sample bias in this data-generating process: here OLS is consistent for β^* but not exactly unbiased at $n = 200$. The slope biases are within about two Monte Carlo standard errors of zero.

Standard errors. The error around the projection, $Y - X^\top \beta^* = \varepsilon + \{m(X) - X^\top \beta^*\}$, combines the noise ε (variance 0.25) with the approximation error of the nonlinear mean, and the second part makes its variance depend on X . For x_1 and x_4 the HC0 RMVs (0.229, 0.251) match the MC SDs (0.230, 0.253) closely, while the model-based RMVs are too large for x_1 (0.238) and too small for x_4 (0.237). This pattern is

consistent with heteroskedasticity; the direction of the model-based error can differ across coefficients. For x_2 and x_3 , both estimators understate the MC SD by up to about 4%, with HC0 closer.

Coverage. A coverage proportion π has Monte Carlo standard error $\sqrt{\pi(1-\pi)/2000}$, which is about 0.005 at $\pi = 0.95$ and 0.007 at $\pi = 0.90$. So coverage near 0.93 is genuinely below nominal. Both methods undercover slightly for the intercept, x_2 , x_3 , and x_4 : HC0 has no small-sample correction, and the intercept is also biased. For x_1 the model-based intervals overcover because its SE is too large, while the HC0 intervals are close to nominal.

The R code for the simulation is

```

1 summarize_mc <- function(B, V_model, V_robust, target) {
2   z90 <- qnorm(.95)
3   z95 <- qnorm(.975)
4   coverage <- function(V, z) {
5     colMeans(abs(sweep(B, 2, target, "-")) <= z * sqrt(V))
6   }
7
8   data.frame(
9     target = target,
10    mean = colMeans(B),
11    bias = colMeans(B) - target,
12    mc_sd = apply(B, 2, sd),
13    model_root_mean_var = sqrt(colMeans(V_model)),
14    model_cov90 = coverage(V_model, z90),
15    model_cov95 = coverage(V_model, z95),
16    robust_root_mean_var = sqrt(colMeans(V_robust)),
17    robust_cov90 = coverage(V_robust, z90),
18    robust_cov95 = coverage(V_robust, z95))
19 }
20
21 set.seed(0)
22 m <- 2000
23 coef_names <- c("(Intercept)", paste0("x", 1:4))
24 a <- c(0, 1, 1, 1)
25 vS <- drop(t(a) %*% msig %*% a)
26 lambda <- sqrt(vS) * sqrt(2 / pi) / 3
27 target_random <- c(vS / 6, 1, rep(lambda, 3))
28
29 B <- V_model <- V_robust <- matrix(NA_real_, m, 5)
30 for (s in seq_len(m)) {
31   x_s <- mvrnorm(n, mu, msig)
32   y_s <- x_s[, 1] +
33     pmax(0, rowSums(x_s[, 2:4, drop = FALSE]))^2 / 3 +

```

```

34   rnorm(n, 0, sig)
35
36   fit_s <- ols_stats(x_s, y_s)
37   B[s, ] <- fit_s$beta
38   V_model[s, ] <- diag(fit_s$V_model)
39   V_robust[s, ] <- diag(fit_s$V_robust)
40 }
41
42 random_results <- summarize_mc(
43   B, V_model, V_robust, target_random)
44 rownames(random_results) <- coef_names
45 random_results

```

7. **Exercise 1.10.** Use the setting of Exercise 1.8, but hold the generated x -matrix fixed.
- Find analytical expressions and numerical values for all fixed-design target coefficients.
 - Using 2000 repeated simulations with x fixed, report the Monte Carlo biases and standard deviations of the five estimated coefficients.
 - Report the square roots of the Monte Carlo means of model-based variances and the 90% and 95% coverage proportions.
 - Report the corresponding model-robust quantities and summarize the simulation results.

Solution: (a) Fix the x generated right after `set.seed(0)` (the same x as in Exercise 1.8) and let $X = (\mathbf{1}, x)$. Given X , we have $Y = m + \varepsilon$ with

$$m_i = x_{i1} + \frac{\{\max(0, x_{i2} + x_{i3} + x_{i4})\}^2}{3}, \quad \varepsilon \sim N(0, \sigma^2 I_n), \quad \sigma = 0.5.$$

So

$$\hat{\beta} = (X^\top X)^{-1} X^\top m + (X^\top X)^{-1} X^\top \varepsilon,$$

and the target is $\beta_X^* = \mathbb{E}(\hat{\beta} \mid X) = (X^\top X)^{-1} X^\top m$, the projection of the 200 true means onto this particular design:

Coefficient	Fixed-design target
Intercept	0.983946
x_1	0.650521
x_2	0.815593
x_3	0.103363
x_4	0.973887

These differ from the random-design targets because they depend on the realized x .

(b)–(c) Model-based results (2000 replications after `set.seed(0)` and the draw of x). Only ε is redrawn in each replication, and coverage is measured against β_X^* .

Coefficient	Target	Bias	MC SD	Model RMV	Cov. 90%	Cov. 95%
Intercept	0.983946	0.000224	0.035434	0.097466	1.0000	1.0000
x_1	0.650521	-0.001225	0.059715	0.161769	1.0000	1.0000
x_2	0.815593	0.000341	0.075042	0.207115	1.0000	1.0000
x_3	0.103363	0.000391	0.078367	0.217560	1.0000	1.0000
x_4	0.973887	0.000558	0.067411	0.181247	1.0000	1.0000

(d) HC0 results (same replications):

Coefficient	HC0 RMV	Cov. 90%	Cov. 95%
Intercept	0.096745	1.0000	1.0000
x_1	0.139671	0.9995	1.0000
x_2	0.229211	1.0000	1.0000
x_3	0.225230	1.0000	1.0000
x_4	0.238594	1.0000	1.0000

Given X , OLS is exactly unbiased for β_X^* , and by the decomposition in (a),

$$\text{Var}(\hat{\beta} \mid X) = (X^\top X)^{-1} X^\top (\sigma^2 I) X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1} = 0.25 (X^\top X)^{-1},$$

which is what the MC SDs estimate. Three quantities should be kept apart here: the noise ε , with variance 0.25; the approximation error $m - X\beta_X^*$ of the best linear fit, which is fixed once X is fixed; and the OLS residual $\hat{r}$, which contains both. The variance estimators are built from $\hat{r}$. With $H = X(X^\top X)^{-1}X^\top$,

$$\hat{r} = (I - H)m + (I - H)\varepsilon.$$

Because the mean is nonlinear, $m \notin \mathcal{C}(X)$, so the first term, $(I - H)m = m - X\beta_X^*$, is a fixed, nonzero approximation error. It inflates both estimators: $\mathbb{E}(\hat{\sigma}^2 \mid X) = \sigma^2 + \|(I - H)m\|_2^2 / (n - k)$, and $\mathbb{E}(\hat{r}_i^2 \mid X) = \{((I - H)m)_i\}^2 + \sigma^2(1 - h_{ii})$. Both treat the fixed approximation error as if it were noise, so the model RMV is approximately 2.7 times the MC SD for each coefficient and coverage is essentially 100%.

This does not contradict Exercise 1.9. Under random design the approximation error changes from sample to sample and really is part of the variability of $\hat{\beta}$; holding one design fixed removes it.

The R code is

```

1  set.seed(0)
2  x_fixed <- mvrnorm(n, mu, msig)
3  X_fixed <- cbind(1, x_fixed)
4  mean_y_fixed <- x_fixed[, 1] +
5    pmax(0, rowSums(x_fixed[, 2:4, drop = FALSE]))^2 / 3
6
7  target_fixed <- drop(
8    solve(crossprod(X_fixed),
9          crossprod(X_fixed, mean_y_fixed)))
10
11 m <- 2000
12 B <- V_model <- V_robust <- matrix(NA_real_, m, 5)
13 for (s in seq_len(m)) {
14   y_s <- mean_y_fixed + rnorm(n, 0, sig)
15   fit_s <- ols_stats(x_fixed, y_s)
16
17   B[s, ] <- fit_s$beta
18   V_model[s, ] <- diag(fit_s$V_model)
19   V_robust[s, ] <- diag(fit_s$V_robust)
20 }
21
22 fixed_results <- summarize_mc(
23   B, V_model, V_robust, target_fixed)
24 rownames(fixed_results) <- coef_names
25
26 target_fixed
27 fixed_results

```